

FACULTY OF ENGINEERING AND BASIC SCIENCES
ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

COURSE: ETL (G01)
ETL-Project: Final Delivery

1. Objectives

The final delivery integrates the concepts and tools developed throughout the ETL course. Students must refine and complete their previous deliveries by implementing a reproducible, automated, and validated data pipeline that combines batch processing, data quality validation, dimensional modeling, analytical reporting, and a streaming component using Kafka.

The final solution must not only demonstrate technical implementation, but also explicitly show how the proposed architecture, analytical outputs, and streaming components support the business objectives defined during the previous project deliveries.

By the end of this delivery, each team must demonstrate a pipeline capable of:

- Extracting data from the original project dataset and one public API selected in the second delivery.
- Orchestrating the batch ETL workflow using Apache Airflow.
- Cleaning, integrating, validating, and loading data into a relational Data Warehouse.
- Using Great Expectations as an automated quality gate before loading analytical tables.
- Maintaining or refining the dimensional model defined in previous deliveries.
- Generating business KPIs and static dashboards from the Data Warehouse, not from CSV files.
- Publishing selected metrics derived from the Fact table to Kafka.
- Consuming streaming metrics and using them for a simple real-time monitoring output.
- Demonstrating how the final ETL platform addresses the original business objectives through analytical insights, KPIs, dashboards, and streaming metrics.

2. Scope and Continuity with Previous Deliveries

This final delivery must build upon the work developed in the first and second deliveries. Teams must reuse the original business problem, the selected static dataset, the public API, and the dimensional model, unless a change is technically justified.

- First delivery foundation: business objective, profiling, dimensional model, ETL code, Data Warehouse, and analytical dashboard.
- Second delivery foundation: API integration, Airflow orchestration, and Great Expectations validation.
- Final delivery extension: end-to-end integration, reproducibility, Kafka streaming metrics, and final technical documentation.

3. What is Expected

3.1 Refined Requirements and Analytical Objectives

Teams must refine the project objectives based on the feedback received in previous deliveries.

- Restate the business problem and analytical goal.
- Explain how the API source improves or enriches the original analysis.
- Define the final KPIs and explain how they support decision-making.
- Explain how success will be evaluated from both technical and analytical perspectives.

3.2 Data Sources

Use the data sources used in the previous deliveries (dataset and API) unless you explicitly justify the use of new sources.

3.3 Exploratory Data Analysis and Data Profiling (Refined)

- Profile the original dataset only if changes were made after the first delivery.
- Profile the API response structure and quality.
- Analyze integration keys, missing values, duplicates, data types, outliers, and inconsistencies.
- Document on how profiling findings influenced cleaning, transformation, validation, and modeling decisions.

3.4 Batch ETL Pipeline with Apache Airflow

Refine the Apache Airflow DAG that automates the batch ETL process (if needed).

- Extract data from the original dataset and the API.
- Clean and standardize both sources.
- Integrate sources according to the defined business logic and grain.
- Transform the data into the structures required by the dimensional model.
- Run Great Expectations validation before loading the final analytical tables.
- Load dimension tables first and fact tables after dimensions.
- Ensure referential integrity and reproducible execution.

The DAG must clearly separate extract, transform, validate, and load stages. It must not be a single monolithic task.

3.5 Data Quality Validation with Great Expectations (Refined)

Great Expectations must be integrated into the Airflow DAG as a validation task.

- Validation must be applied after transformation and before loading the final dimensional model.
- Validation must reflect business rules, data model constraints, and grain consistency.
- Critical validation failures must stop the load process.

- Non-critical issues may be logged but must be documented.
- Teams must provide the expectation suite and validation results.

Recommended expectations include:

- No null values in required keys and measures.
- Uniqueness of the fact table grain.
- Valid ranges for numerical measures.
- Accepted values for categorical attributes.
- Referential integrity checks where applicable.
- Freshness or timeliness checks for API data when relevant.

3.6 Dimensional Model and Data Warehouse (Refined)

Teams must preserve, refine, or extend the dimensional model developed in previous deliveries.

- Define or confirm the grain of the fact table.
- Clearly identify fact table(s), dimension tables, measures, and attributes.
- Use surrogate keys in dimensions.
- Use explicit primary keys and foreign keys.
- Explain any changes made to the previous model and justify why they were necessary.
- Load the final model into a relational Data Warehouse.

3.7 Analytical Dashboard (Refined)

Build a static analytical dashboard using the Data Warehouse as the source.

- The dashboard must connect to the database, not to CSV or local processed files.
- It must include business-relevant KPIs aligned with the refined objectives.
- It must include visualizations that support decision-making.
- Each visualization must have a short interpretation or insight.

3.8 Kafka Streaming Component

Implement a streaming component that publishes metrics derived from the Fact table.

- The Kafka producer must read at least three metrics derived from the Fact table or analytical tables.
- The producer must continuously publish these metrics to a Kafka topic.
- The Kafka consumer must subscribe to the topic, receive the metrics, and display or persist them in real time.
- The streaming component must not read directly from the original CSV file.
- The selected metrics must be meaningful for monitoring the business process or analytical outcome.

3.9 Business Objectives Achievement

Each Team must explicitly demonstrate how the final ETL platform addresses the business objectives defined in the project.

The final solution must show how:

- the ETL pipeline,
- the validation process,
- the dimensional model,
- the dashboard,
- and the streaming component

support the proposed business goals and decision-making process.

Required Analysis

Each Team must include a final section titled: Business Objectives Achievement

This section must:

- Restate the original business objectives.
- Explain which KPIs or analytical outputs support each objective.
- Demonstrate how the dashboard and/or streaming analytics answer the proposed business questions.
- Explain how the ETL architecture supports decision-making.
- Discuss limitations and possible future improvements.

Important

Each KPI, dashboard component, or streaming metric must be explicitly linked to at least one business objective proposed in the project.

Suggested table:

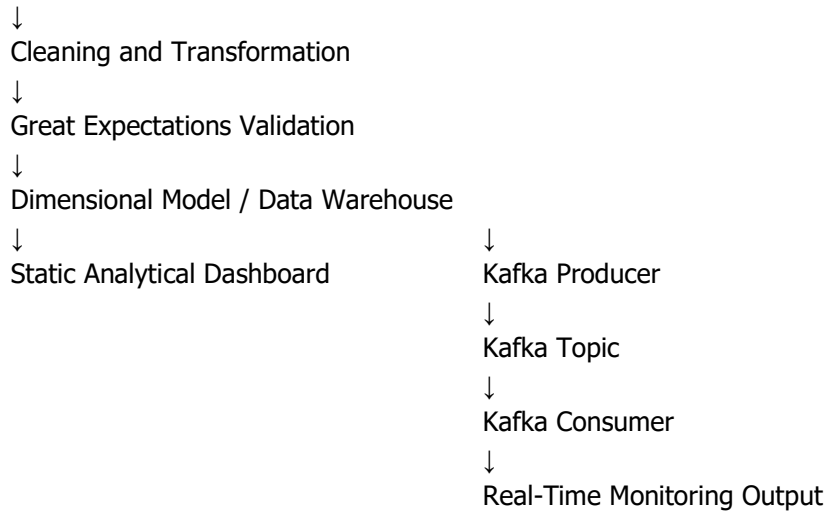
Business Objective	Analytical Question	KPI / Metric	Dashboard or Streaming Evidence	Decision Supported
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The project will not be evaluated only on technical implementation, but also on its ability to generate analytical value and support decision-making.

4. Proposed Architecture

The final solution should follow this general architecture:

Dataset + API
↓
Airflow Batch ETL



5. Technology Stack

Layer	Required / Suggested Tool	Purpose
Orchestration	Apache Airflow	Automate the batch ETL workflow
Data Sources	Original dataset + Public API	Provide static and dynamic data
Transformation	Python / Pandas	Cleaning, integration, feature engineering, and model preparation
Data Quality	Great Expectations	Automated validation before loading final analytical tables
Data Storage	PostgreSQL / MySQL	Persist dimensional model and analytical tables
Streaming	Apache Kafka	Publish and consume selected metrics derived from the Fact table
Visualization	Power BI / Tableau / Looker Studio / Streamlit / Python	Static dashboard and real-time monitoring output

6. Deliverables

6.1 GitHub Repository

- Airflow DAG(s).
- ETL scripts organized by responsibility.
- Great Expectations configuration and expectation suites.
- SQL scripts to create and load the Data Warehouse tables.
- Kafka producer and consumer scripts.
- Dashboard files, screenshots, or links.
- Updated architecture diagram.
- requirements.txt or Dockerfile.

- .gitignore file.
- README.md with clear execution instructions.

6.2 README Requirements

1. Refined business objective and analytical questions.
2. Architecture diagram.
3. Description of both data sources and justification of the API.
4. Summary of EDA and profiling findings.
5. Data integration strategy and assumptions.
6. Dimensional model description, including grain.
7. Great Expectations validation strategy and results.
8. Airflow DAG design and task dependency explanation.
9. Kafka streaming design and selected metrics.
10. Static dashboard results and business insights.
11. Real-time monitoring output and interpretation.
12. Challenges, limitations, and possible future improvements.
13. Setup instructions.
14. How to run the Airflow DAG.
15. How to run the Kafka producer and consumer.

7. Suggested Repository Structure

```
project/
├── dags/
│   └── final_etl_dag.py
├── data/
│   ├── raw/
│   └── processed/
├── src/
│   ├── extract.py
│   ├── transform.py
│   ├── validate.py
│   ├── load_dw.py
│   └── utils.py
├── great_expectations/
├── kafka/
│   ├── producer_metrics.py
│   └── consumer_metrics.py
├── sql/
│   ├── create_tables.sql
│   ├── load_tables.sql
│   └── kpi_queries.sql
└── dashboards/
```

```

|— reports/
|   |— final_report.pdf
|
|— README.md
|— requirements.txt
|— .gitignore

```

8. Evaluation Rubric

Criteria	Weight
Data Integration & Cleaning	0.7
Airflow ETL Pipeline	0.7
Data Validation	0.6
Dimensional Modeling	0.6
Kafka Streaming Integration	0.6
Real-Time Metrics	0.4
Dashboard, KPIs & Business Insights	0.5
Business Objectives Achievement	0.6
Documentation & Reproducibility	0.3

9. Final Note

This final delivery must demonstrate not only that the tools work, but that the team can design a reliable, explainable, and reproducible data engineering solution. The strongest projects will clearly connect business objectives, data modeling, validation, orchestration, streaming, and visualization into one coherent system.